

What is farm productivity?

Productivity is a key measure in the statistical toolbox used to assess farm industry performance.

It is typically defined as a ratio of outputs produced (such as crops, livestock and wool) to inputs used (such as land, labour, capital, materials and services), and productivity growth is a change in this ratio over time.

If farmers produce the same outputs using fewer inputs or more output with relatively fewer inputs, then productivity goes up.

This can occur within an individual farm if an improved production system is used, or it can happen at the industry level if resources such as land and labour move to more productive managers. In this sense, productivity is simply a measure of how efficiently farms or industries are transforming inputs into outputs.

Why productivity matters

Productivity is a popular statistical tool used to check the strength and 'health' of economic growth. It is a measure of performance derived from the observed input and output choices of firms or industries.

Productivity estimates are widely used and compiled for many sectors of the economy by statistical agencies around the world, as well as by international agencies such as the OECD. Within Australia, the Australian Bureau of Statistics (ABS), Productivity Commission (PC), Australian Bureau of Agricultural and Resource Economics and Sciences (ABARES) and various universities all produce and analyse productivity statistics.

In the context of Australian agriculture, increasing productivity has long been recognised as the most important source of long-term growth in farm profits. For Australian farmers, it has been reflected in an increase in the production of outputs relative to inputs, lower production costs and higher incomes. Productivity growth also enables farmers to earn higher returns on capital, higher wages, higher profits and can increase the agricultural contribution to national income. It can also lead to lower prices for consumers and reduced environmental impacts.

Productivity growth is essential for maintaining and improving international competitiveness and has been a means of offsetting declining real prices received for farm commodities on global markets. This is particularly important for Australian farmers, because an average of 70% of the value of Australian agricultural products are exported. Productivity growth in agriculture can also mitigate the adverse effect of other long-term challenges such as population ageing, increased competition for resources such as land and water and climate change.

Productivity or profit

Most farmers are focussed on profit rather than productivity, but the two are closely linked. Profit is determined by the quantities of inputs used, the outputs produced, and by prices paid and received. Farmers generally cannot control the input or output prices, however, they can control changes to their farm business. By improving farm efficiency through technology adoption, better management practises, innovating and so on, farmers can produce a greater quantity of output from their inputs, increasing farm productivity. Assuming constant prices, this productivity increase would likely translate into higher farm profit.

How ABARES measures productivity

ABARES annually releases aggregate productivity estimates for the cropping, mixed livestock-cropping, beef, sheep, and dairy industries.

These productivity estimates are based on data collected through the Australian Agricultural and Grazing Industries Survey (AAGIS) and the Australian Dairy Industry Survey (ADIS). These surveys are completed annually by approximately 1,600 broadacre farmers and 300 dairy farmers. Farms are selected from the business population maintained by the ABS and assigned weights so that their data is representative of the farm population at regional, state and national levels.

ABARES uses a conventional growth accounting approach to estimate productivity. This approach aligns with the OECD Manual on productivity measurement and is broadly in line with the productivity estimation method used by the ABS. Given the nature of the AAGIS and ADIS data, the ABARES technique has unique advantages. For example, it has enabled ABARES to compile aggregate productivity statistics at the national, industry and regional levels and for individual farms, and to construct sophisticated 'climate-adjusted' productivity estimates. In addition, the farm level productivity estimates and the supplementary information in the survey data make it possible for ABARES to undertake detailed econometric and statistical analysis to provide further economic and policy insights which are otherwise impossible.

To estimate farm productivity, ABARES follows a 3-step process (see figure below). In step 1, the capital input is constructed by aggregating machinery, buildings, and other productive capital items. Outputs and other inputs (labour, land, materials, and services) are constructed in a similar manner. In step 2, the total output and total input variables are constructed by aggregating the respective outputs and inputs using prices as weights. Finally, in step 3, the productivity index is generated as the total output index divided by the total input index. The 'aggregation' of farm level data at the various steps in Figure 1 is done using the Fisher index formula.

‘Climate adjusted’ productivity versus ‘normal’ productivity.

‘Climate adjusted’ productivity is basically how productivity would look without fluctuations in short term weather and long-term climate – in other words, average long-term weather every year.

This is achieved using a machine learning simulation model [FarmPredict](#) developed by ABARES.

The purpose of these ‘climate adjusted’ productivity estimates are to observe the real underlying improvements through technological progress, R&D investment, innovation, management skill and so on. Since farm productivity is highly sensitive to seasonal weather variability, real improvements can be difficult to observe, particularly in the short-term. Once climate effects are removed from the estimates, real underlying improvements in farm performance can be observed more clearly.

‘Normal’ productivity on the other hand, is a real-world estimation of productivity using unadjusted farm survey data. It represents the actual broadacre farm population, completely exposed to all climate conditions.

Which productivity estimate to use

Both ‘climate adjusted’ and ‘normal’ productivity have their own purposes. As a general guide, we recommend they are used accordingly:

‘Normal’ productivity

- Requirement is for real-world productivity estimates.
- Use for comparisons to other productivity estimates (e.g., international comparisons)
- Observations of long-term growth

‘Climate adjusted’ productivity

- Requirement is for simulated or synthetic estimates with the effect of weather removed
- Suitable for observing underlying farm productivity, and may be helpful to policy makers (e.g., wanting to assess the benefits of R&D investment)
- Useful for making short-term growth observations since much of the short-term volatility is removed.

Source: <https://www.agriculture.gov.au/abares/research-topics/productivity/productivity-introduction>